Leadership & Consulting in Data Science

Week 5



McDonald's Presentation - Epilogue

For this project, the team built a times series forecast model executed at restaurant level to evaluate incremental traffic and profitability across menu items. In general, the model showed that the transaction decline was interrupted by the "All Day Breakfast" promotion. Therefore, the entire breakfast menu was offered all day and continued until the COVID 19 pandemic, when it was suspended. A good portion of the McDonald's breakfast customers were coming for breakfast items later in the day while regular lunch customers continued to buy regular lunch items (e.g., burgers). So, All Day Breakfast was not cannibalizing higher profit lunch items and brought McDonald's incremental traffic to the restaurant throughout the day.

McDonald's Case Debrief

- **Lift.** Need an appropriate way to answer Regional Manager's question. How do we create a reasonable measure of lift? Statistically significant?
- **Aggregate vs. Item level data.** Take care not to mix information.
- **Missing Data.** New stores opening? Stores closed for renovation? How to deal with those stores…exclude, impute, use as is?
- **Outliers.** High / low volume stores. Single events. High # of items per transaction…work environments where single person sent to pick up orders for many.
- **Seasonality.** When was the test done? Locals vs. Tourists. Does this give a good read?
- **Compliance.** Did all stores run the promotion and in the same way?
- **Enough Equipment.** Was there enough equipment in stores to see what happens? Check busy stores. Check stores with Chicken drop.
- **Advertising / Promotion.** What did McDonald's do to advertise this promotion? Not clear from info. Example, if local paper FSI, then tourists would be less likely to be aware of the promotion. What would it be like if this was nationally advertised? In general, could promotion be store focused when so many, so close together…customer expectations
- **Price.** While seeing the 5% / year there was also a price increase every year. Was this consistent with inflation / cost of living? Could this be supply/demand issue.
- **Profits.** Can you find information on Profitability? Can you use that to approximate stories. Fries much more profitable than Big Mac.

McDonald's After-Action Review



Now is a good time for your team to review your team's performance on McDonald's and identify what you will do for the Verizon presentation.

- What did you do well and want to ***continue*** doing?
- What caused issues and do you want to ***stop*** doing?
- How could you have improved and you want to ***start*** doing?

Think about this across the following areas

- Problem understanding and answer
- Timing, scheduling, and completion (enough time given?)
- Work assignments and roles (everything covered well?)
- Slide writing / QA
- Presenting (sufficient practice?)
- Leadership and coordination

Create a one or two page presentation and be prepared to share it with the class.

Agenda

1



Decision Making

2



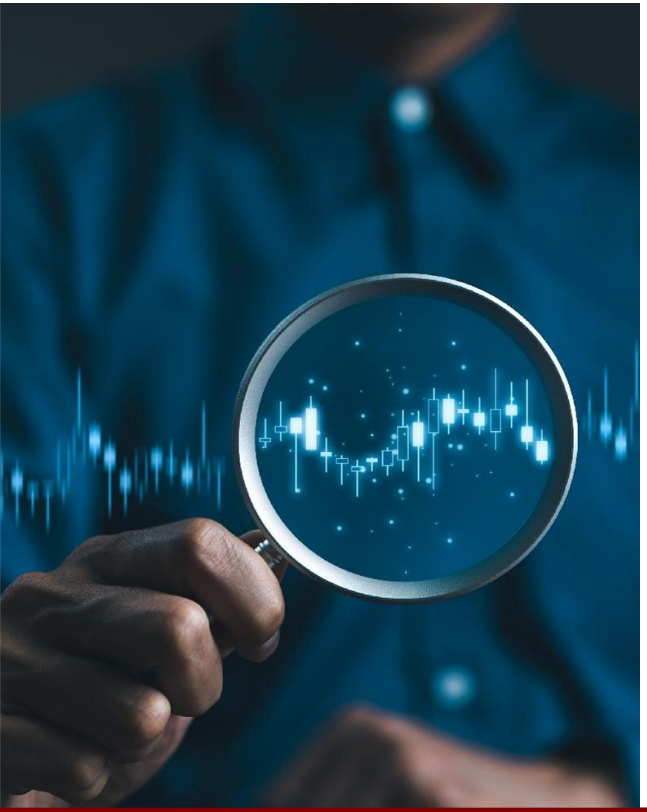
Problem Solving

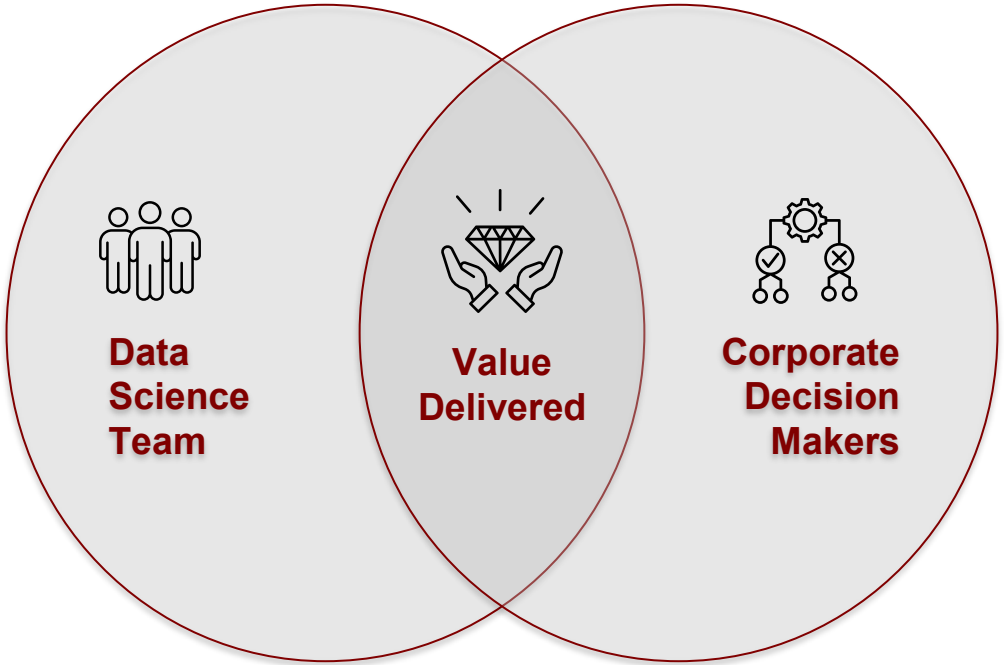
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Creativity

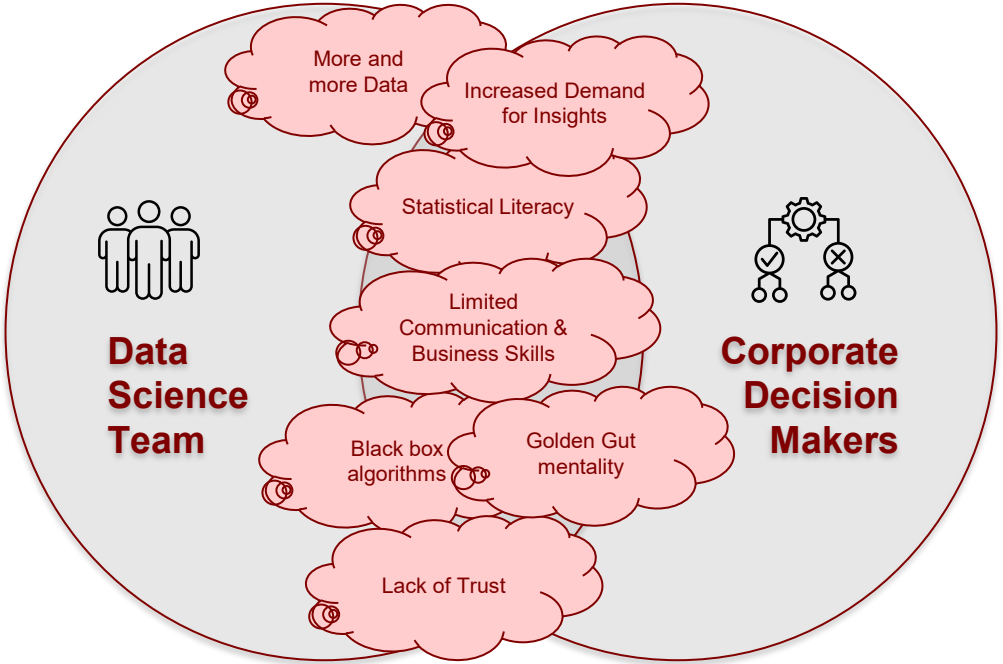
The goal of the Data Science team is to work with Decision Makers and deliver value to the organization



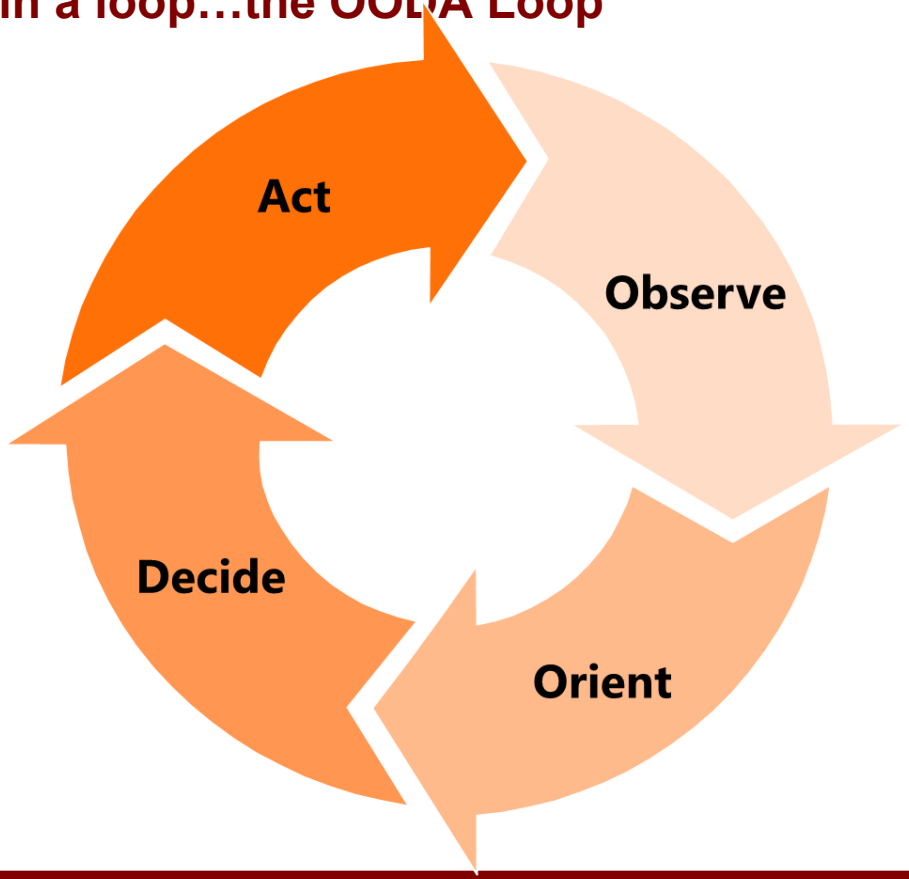


Unfortunately, there's a perfect storm separating the two sides

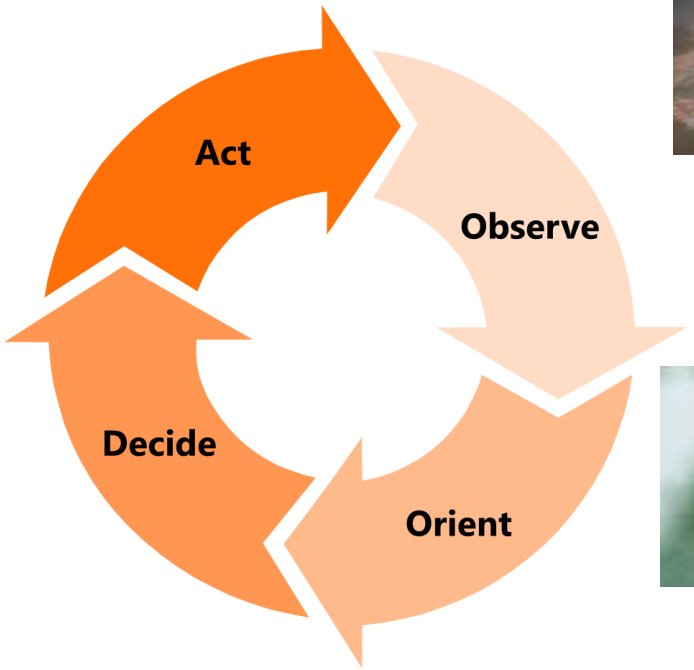




Decisions are made in a loop…the OODA Loop



If you can make decisions faster than the competition, you win!



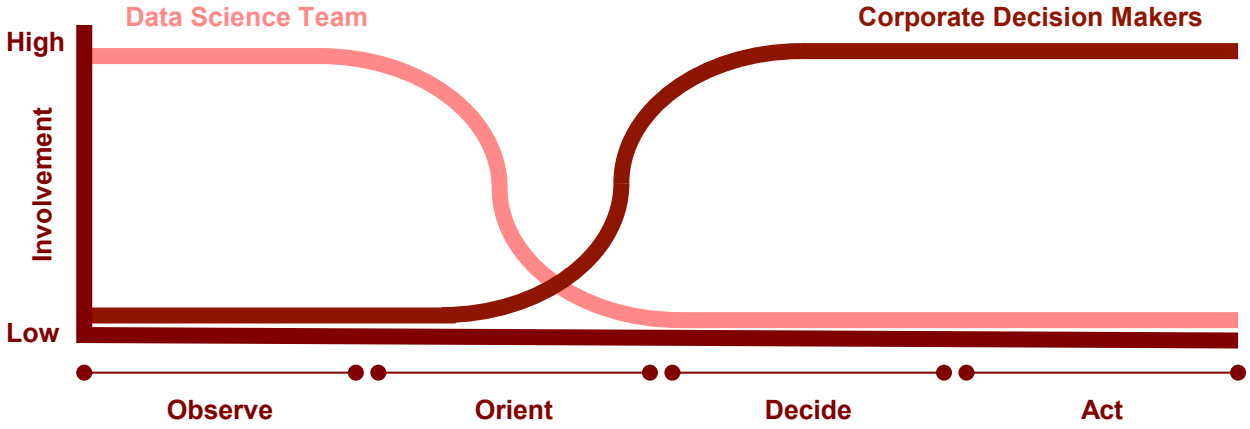


Mind the gap between Observe / Orient and Decide / Act

The 2026 AI & Data Leadership Executive Benchmark Survey identified persistent **obstacles** to data-driven transformation

However, the biggest obstacles were **NOT technical – 93% cited human issues; only 7% cited technology**

The top challenge was **culture and change management**…aligning the organization to **translate analytics into real world action**



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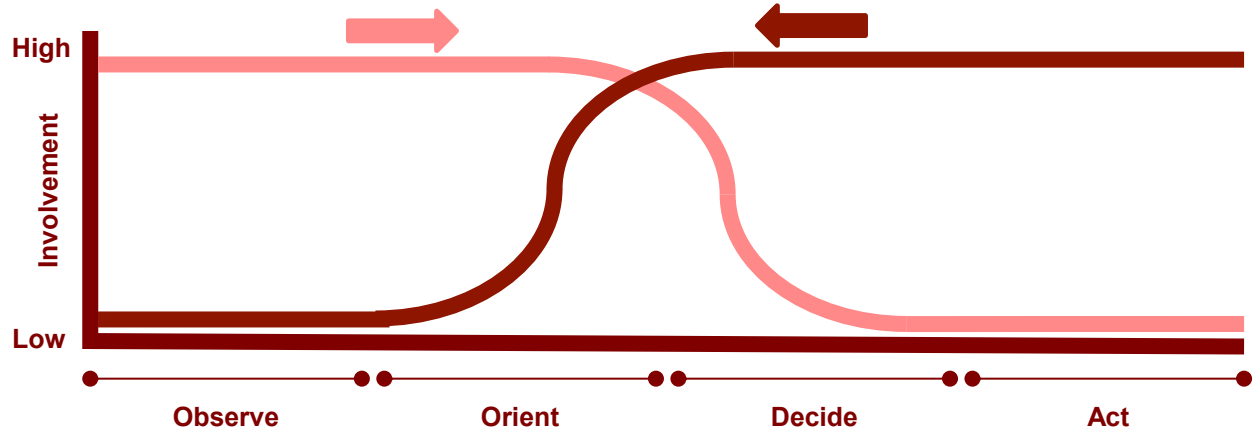
Bridge the gap from either the Data Science Team or Decision Makers

Data Science Team

- Improve core data science capabilities
- Build non-quant skills
- Get involved in the business
- Understand the connection of analytics to real world business solutions

Corporate Decision Makers

- Develop quantitative skills
- Trust the data and drive toward a data-driven decision culture
- Include data science team in the process



Impaired team decision making…groupthink

Teams often have an informal decision-making process—so informal that members do not recall a decision being made

Team decision-making can be impaired by premature closure to avoid disagreement, time pressure, or outside stress; factors such as



Lack of healthy conflict and/or critical analysis



GAGA - Go Along/Get Along



Conformity with the leader



Ignoring input from outside sources



Time pressure leading to desire of team to accept first plausible decision

Teams can take three approaches to making decisions

1

Consultative

An individual holds decision rights and decides after gathering more or less input from the team.

one-way pronouncement, human filing cabinet, processing information with individuals or with the team

2

Democratic

The team decides by vote; a minority or majority carries the decision.

minority rule and majority rule

3

Consensus

Discussion continues until every member can accept and support the decision; unanimity is the strongest form.

consensus and unanimity

Choosing an approach means trading off **quality**, **acceptance**, and **time** — the eight options on the next slide span this continuum.

In general, the greater the time pressure, the more intense the drive to autocratic decision making

1. One Way Pronouncement	2. Human Filing Cabinet	3. Process Info with Individuals	4. Process Info with Team	5. Minority Rule	6. Majority Rule	7. Consensus	8. Unanimity
Likely Lower Quality		←	—	—	→	Likely Higher Quality	
Likely Less Acceptance		←	—	—	→	Likely More Acceptance	
Less Time		←	—	—	→	More Time	



Expediency may sacrifice commitment because it may not allow for those affected to understand and to then get behind a decision and for those who implement to be committed to getting the work done



The time spent getting others committed to an autocratic decision may be greater than the time needed to reach consensus



Urgency is the driver. More urgent: go with autocracy or majority vote. Less urgent: go with consensus, knowing you can always go to majority vote or autocracy if needed

Being expeditious may be dangerous

Teams face problems from the beginning, but are not equipped to deal with them



Problems are ill-defined…

- Dilemma with no apparent solution
- Undesirable situation without a way out
- Questions that cannot be currently answered
- Difference between current state and desired, future state
- Situations team members must manage effectively

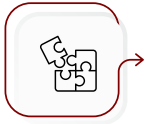


Teams have trouble solving problems…

- They generate solutions before defining and understanding the problem
- The solution recommendation is separated from solution implementation
- No one seriously analyzes the success and/or failure of the solution

"Defining and evaluating the problem is the most difficult step for teams to perform… Teams may rush through the problem definition stage, only to find that they have to return to it during the solution or implementation stage." Levi, Group Dynamics, p. 185

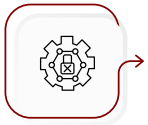
Teams encounter difficulties in problem solving



Applying solutions that worked in the past



Basing decisions on intuitive/emotional thinking rather than rational and logical thinking



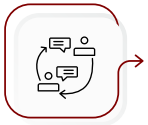
Limited time, money, and information



Finding acceptable but suboptimal solutions



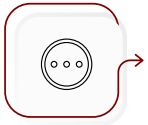
Difficulty determining best solution



Communication issues in the team



Politics!



Other

The problem-solving process has three stages

1

Recognize & define the problem

- Gather data on symptoms across the systems involved
- Separate symptoms from underlying causes (five whys, root cause analysis)
- Agree on the gap between the current state and the desired future state

2

Generate & select solutions

- Diverge first: brainstorm alternatives before judging them
- Converge: evaluate alternatives against agreed criteria (criteria matrix)
- Select the solution the team can commit to

3

Implement & evaluate

- Build an action plan with owners and milestones
- Keep recommenders connected to implementers to avoid disconnects
- Evaluate results against the problem definition and feed learning into the next loop

Problem solving is iterative: evaluation restarts observation — completing the OODA Loop for the next cycle.

Problem solving techniques

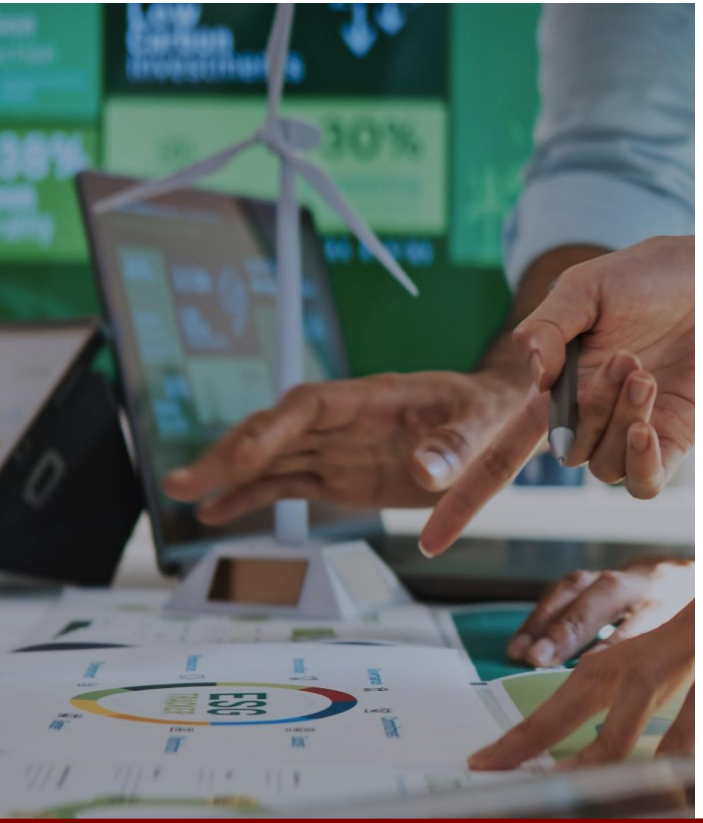


Process Mapping: Map current and future (desired) state

- Helps define problem and avoid jumping immediately to solutions
- Good for temporary teams; problem solving teams
- Good for diverse teams with varying levels and types of communications
- Provides an established structure

 **Problem Analysis**	Separate symptoms from causes; the five whys; root cause analysis
 **Force Field Analysis**	Identify driving forces and restraining forces; more successful if reducing restraining forces from preventing change
 **Criteria Matrix (select solution)**	Rate alternatives; weighted and unweighted decision-making tables
 **Action Plans (improve implementation)**	Strategic plans; milestones

Creativity is needed for teams to solve problems



The search for ideas that are both new and useful

Novelty alone is not sufficient; the idea must also effectively solve a problem

Creativity requires divergent and convergent thinking

- Divergent: Generating potential ideas
- Convergent: Analyzing and focusing on solutions

Creativity requires a set of capabilities



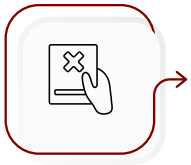
Domain-relevant skills

Task motivation

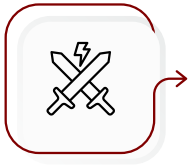
Appropriate cognitive styles

- Ability to break out of established mental sets: viewing something from a new perspective
- Appreciation of complexity
- Suspension of judgment
- Use of broad categories to view issues

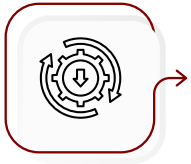
Creativity can be hampered



Negative (or critical) communication climates



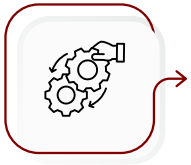
Interpersonal **conflicts** among individuals



Inefficiency: team creativity is slower than individuals working alone



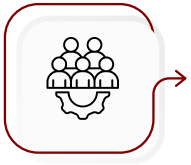
Conformity pressures and dominating members



Cognitive interference: disruption of thinking while one is waiting to speak



Social inhibition: **anxiety** about how others will evaluate you and **loafing** (lack of motivation in creative activities)



Long-term teams without membership change and/or new teams of total or near total strangers may not be as creative

Teams can enhance their creativity

Diversity – Experience, Cultural and Approach

Some degree of familiarity among team members (**trust**)

- Perspective taking: being encouraged to see things the way that someone else would see it (“How would X handle this?”)
- Exposure to other ideas helps individuals break out of limiting cognitive categories
- Incubation periods: taking breaks during the creative process and/or having multiple creativity sessions
- Combining both individual and group creativity

Some degree of dissent or “**devil’s advocacy**”

Teams must create an environment that supports creativity



- Designate non-critical times (no feedback – all ideas welcome)
- Adhere to the ground rules of brainstorming
- Create norms for openness and safety
- Develop the skill and willingness to build on an idea rather than tearing it down
- "Yes and…" vs. "No but…"--- build on the idea being presented before deciding it will not work

Individual Feedback



Get together in groups of two. Practice giving feedback (at least one positive and one negative) using the following:

- "When you respond politely after a teammate insults you…it diffuses the situation…much appreciated!"
- "When you play devil's advocate…it forces us to evaluate our position…thanks!"
- "When you deliver your EDA on time…it keeps us on schedule…keep up the great work!"
- "When you are late to the meeting…it makes everyone stay longer…could you change that?"
- "When you do not deliver a slide on time…we can not QA and practice the presentation…how could you do that better?"

Use the following scenarios to create and deliver ***additional*** feedback:

- A teammate has produced an interesting insight from their EDA
- A teammate found a problem in the model code and was able to fix it
- A teammate's suggestion on a slide helped clean it up and make it more understandable
- A teammate does not practice their part of the presentation, and it ends up being too long during the client delivery

Create you own scenarios from your team or own team experiences

Fix a Slide



Select a slide or slides from your McDonald's presentation that you feel could be improved. Consider the following:

- Precise and concise
- Storyline titles
- Vertical logic
- Visualization
- Legends
- Sources

Prepare a presentation that includes the original slide or slides and the "fixed" slide. Be prepared to present this to the class

End of Deck